

Pattern Recognition and Anomaly Detection

Semester - VI

Industrial Machine Anomaly Detection: A Comparative Analysis

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Problem Statement

- Industrial machines and IoT devices generate vast amounts of time-series data.
- Anomalies in this data can indicate critical issues:
 - System failures
 - Maintenance needs
 - Security breaches
- Early detection can prevent costly downtime, safety hazards, and financial losses.
- Accurate anomaly detection is crucial for proactive maintenance and operational efficiency.

- Unexpected machine failures in industries lead to high costs and safety risks. Detecting anomalies early using time-series sensor data can prevent such issues. This project explores machine learning models to build reliable, interpretable, and proactive anomaly detection systems for smarter industrial maintenance.
- By applying machine learning models, we can develop accurate and interpretable systems that identify potential issues before they escalate. This not only improves operational efficiency but also supports the vision of smart manufacturing.

Literature Review

S.No	Authors	Year	Title	Key Findings	Technology Used	Relevance to Project
1	Sérgio F. Chevtchenko et al.	2023	Anomaly Detection in Industrial Machinery using IoT Devices and Machine Learning: a Systematic Mapping	Comprehensive evaluation of 84 studies from 2016 to 2023, identifying commonly used algorithms, preprocessing techniques, and sensor types in industrial anomaly detection. [1]	Various ML algorithms, IoT sensor data	Overview of techniques and gaps, guiding ML model selection for anomaly detection.
2	Umer et al.	2024	Machine Learning for Anomaly Detection in Industrial Environments	An overview of ML applications in industrial anomaly detection, highlighting challenges, future prospects.[2]	Random Forests, k-NNs, Decision Trees, SVMs, ANNs	Offers insights into model selection and dataset impact on performance.
3	Md Ismail Hossain, Shahinuzzaman Sanim,S. M. Kamruzzaman, Abu Shufian	2023	Anomaly Detection in Industrial Machinery Using Three-Axis Vibration Data	Anomaly detection with vibration data using One-Class SVM, Isolation Forest, and LSTM. [3]	One-Class SVM, Isolation Forest, LSTM Autoencoder	Provides practical ML examples for vibration data, relevant to similar sensor projects.
4	Freeman Paul, Kayode Sheriffdeen	2024	Anomaly Detection in Industrial Sensors: A Machine Learning-Driven Approach to Predictive Maintenance	ML-driven approach for predictive maintenance, highlighting advantages over traditional sensor malfunction methods. [4]	Various ML models for predictive maintenance	Highlights ML benefits in predictive maintenance for early anomaly detection and failure prevention.
5	Frontiers in Artificial Intelligence	2024	Explainable Correlation-Based Anomaly Detection for Industrial Control Systems	Proposes explainable correlation-based anomaly detection using LSTM-AE and Pearson correlation. [5]	LSTM Autoencoder, Pearson Correlation	Emphasizes explainability and efficiency in anomaly detection, important for real-time applications in industrial settings.

Table-1 Literature Review of related papers

Methodology

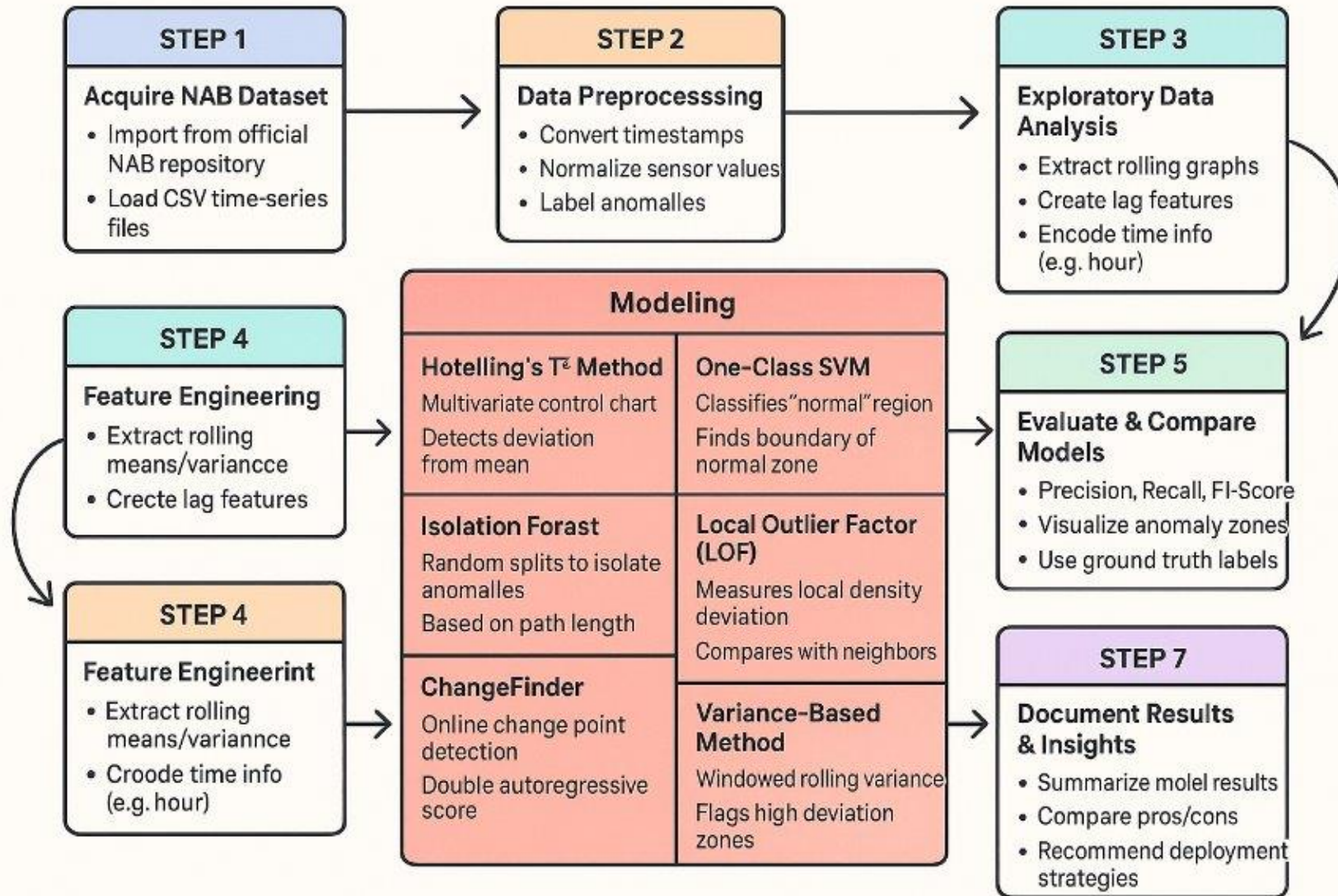


Fig.1 End-to-End Work Flow of Industrial Machine Anomaly Detection

Technology Stack

- **Programming Language:** Python
- **Libraries:**
 - **Data Manipulation:** pandas, numpy
 - **Visualization:** seaborn, matplotlib
 - **Anomaly Detection:** scikit-learn, ChangeFinder, Hotelling's T^2
 - **Model Explainability:** SHAP
- **Development Environment:** Jupyter Notebook, Google Colab
- **Operating System:** Windows, Mac
- **Version control:** Git & GitHub for source code management

Dataset Description

Attribute	Description
Dataset Source	Numenta Anomaly Benchmark (NAB) [6]
Domain	Time-series anomaly detection
Format	CSV file with timestamped sensor readings
Data Type	Real-world machine sensor data (temperature readings)
Timestamp Column	timestamp – Records the time of each sensor reading
Value Column	value – Temperature of the machine over time
Anomaly Labels	Known failure period implied by timestamps (available in separate label files)
	Around 21,600 rows (~15 days of data)

Table-2 NAB Dataset Description

	timestamp	value
0	2013-12-02 21:15:00	73.967322
1	2013-12-02 21:20:00	74.935882
2	2013-12-02 21:25:00	76.124162
3	2013-12-02 21:30:00	78.140707
4	2013-12-02 21:35:00	79.329836

Fig.2 machine_temperature.csv

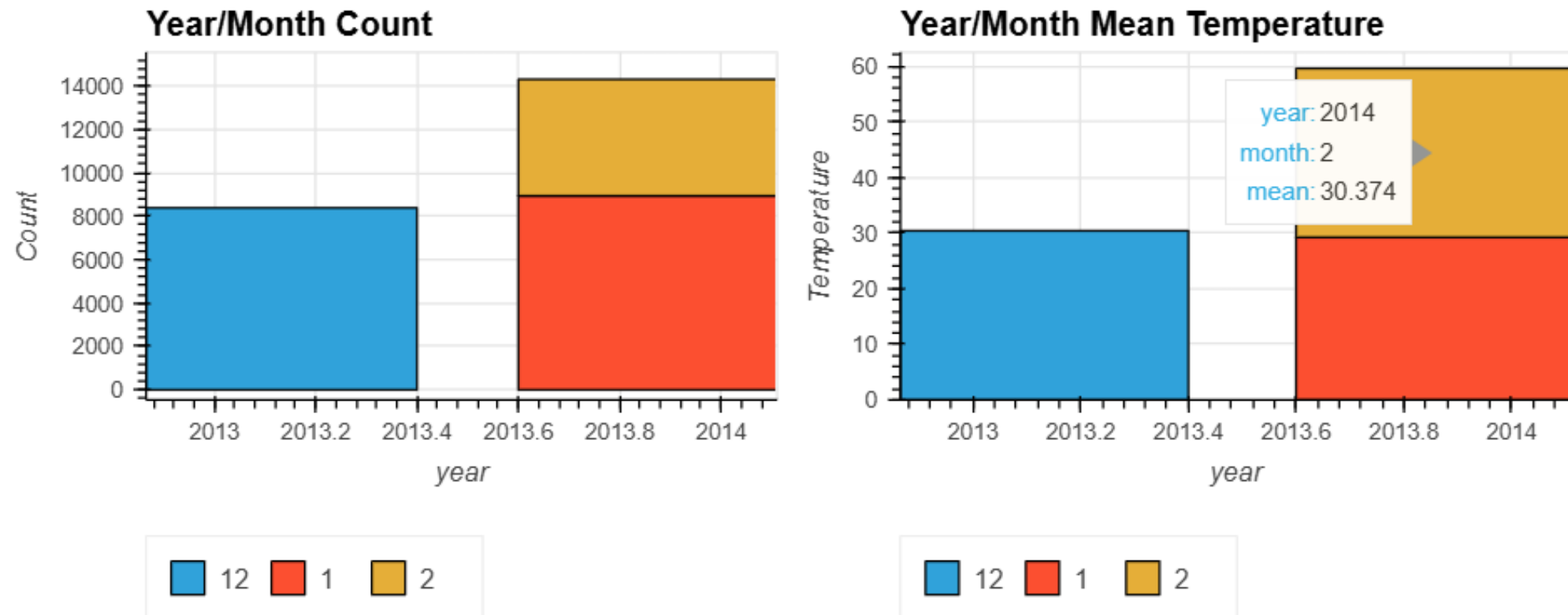


Fig.3 Mean temp. Data across different years

- These graphs presents a comparison of count and mean temperature data across different year/month periods.

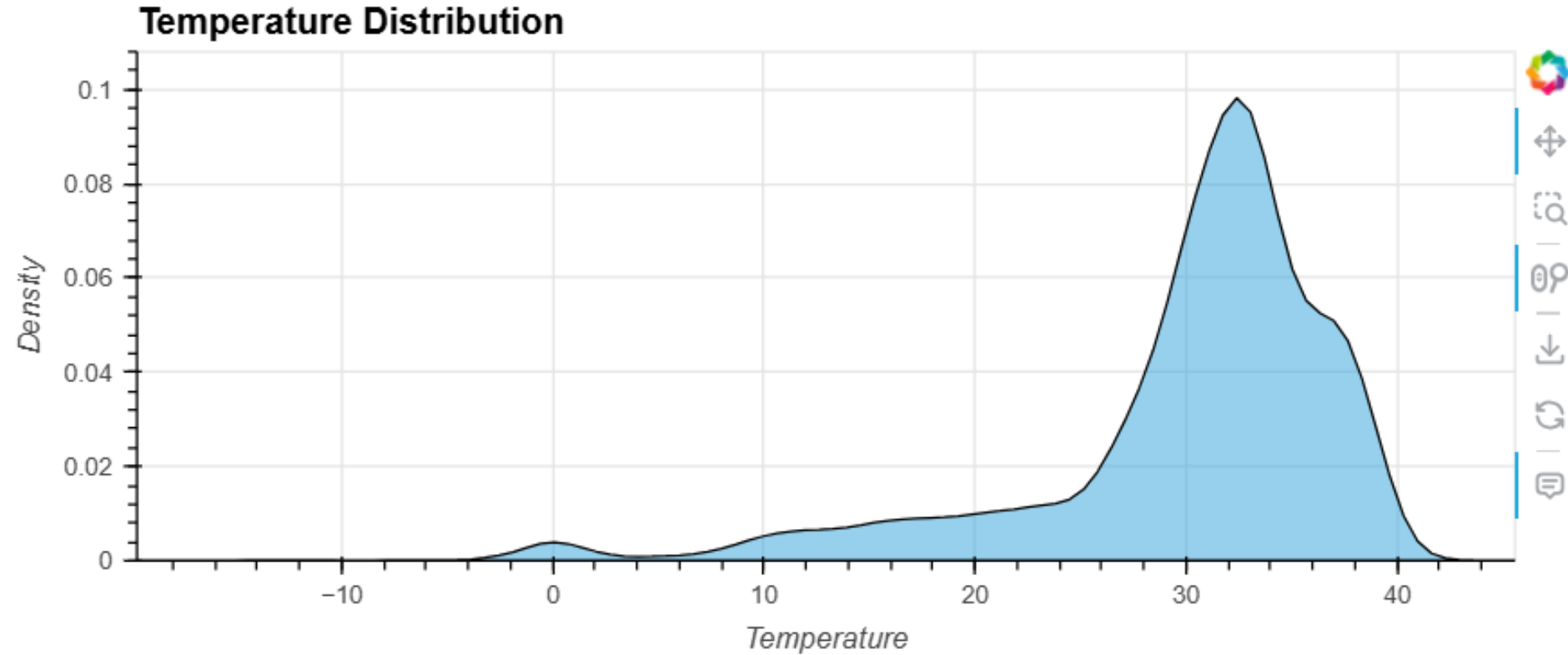


Fig.4 Temperature Density

- The plot suggests that, temperatures around 30 degrees are considered "normal" since that's where the density is highest.

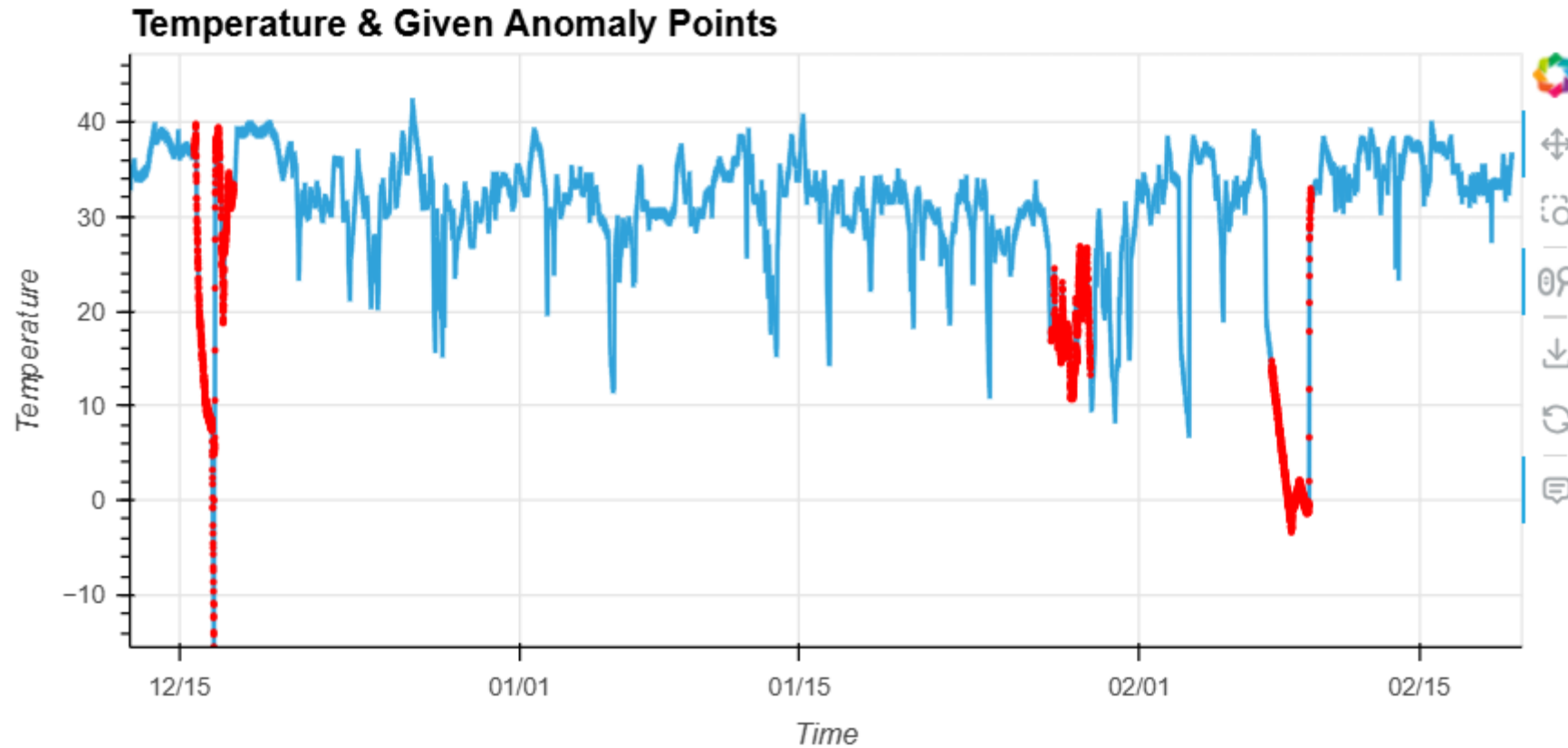


Fig.5 Temp-time series and anomaly points

- This plot shows where the known anomalies are located within the temperature time series.

Data Preprocessing

STEP	DESCRIPTION
1. Load Dataset	Imported NAB time-series CSV files (sensor readings with timestamps).
2. Timestamp Conversion	Converted string-based timestamps to datetime objects for time-based analysis.
3. Sorting by Time	Ensured all records are ordered chronologically.
4. Missing Value Handling	Checked for and handled any missing or null values (e.g., with interpolation or forward fill).
5. Label Alignment	Loaded ground truth anomaly labels and matched them with the corresponding timestamps.

Table-3.1 Data Preprocessing

Data Preprocessing

STEP	DESCRIPTION
6. Label Encoding	Created binary anomaly labels (1 = anomaly, 0 = normal) for supervised evaluation.
7. Normalization / Scaling	Applied Min-Max Scaling or Standardization to the sensor values.
8. Rolling Statistics	Calculated moving averages and variance over a sliding time window.
9. Feature Engineering Prep	Prepared dataset for feature extraction (e.g., lag values, statistical summaries).
10. Data Subsetting	Focused on specific time windows or datasets for experimentation.

Table-3.2 Data Preprocessing

Feature Engineering

Different Clusters with the 2 Main Features

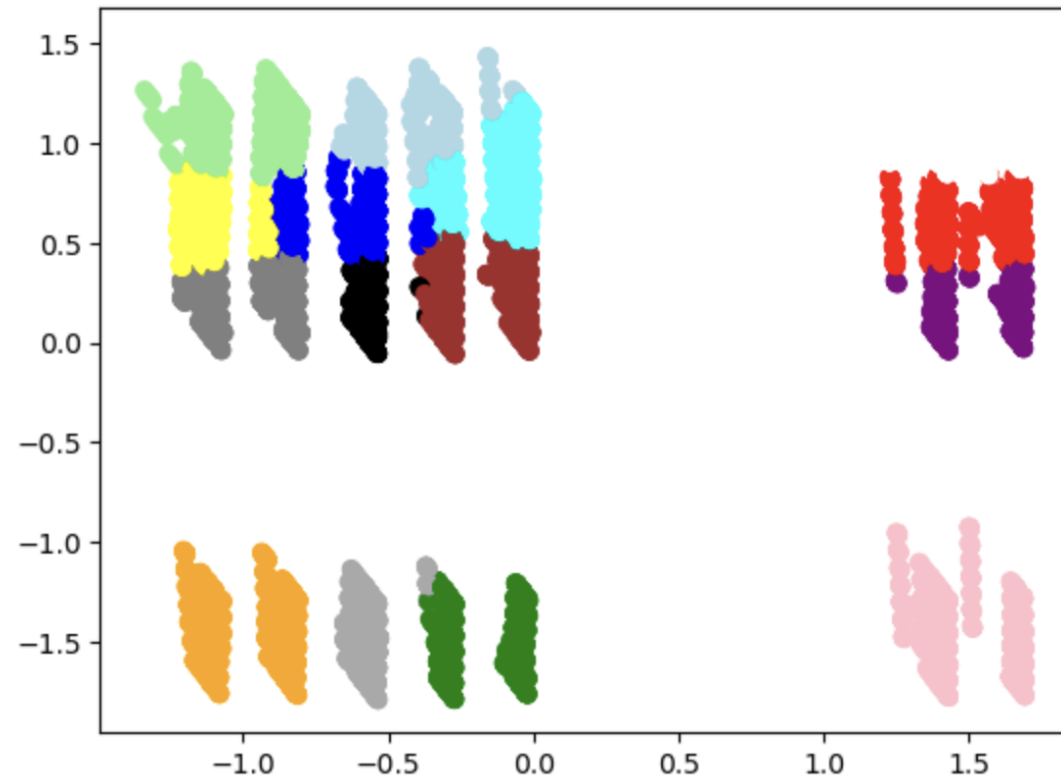


Fig.6 Clustering of 2 PCA features

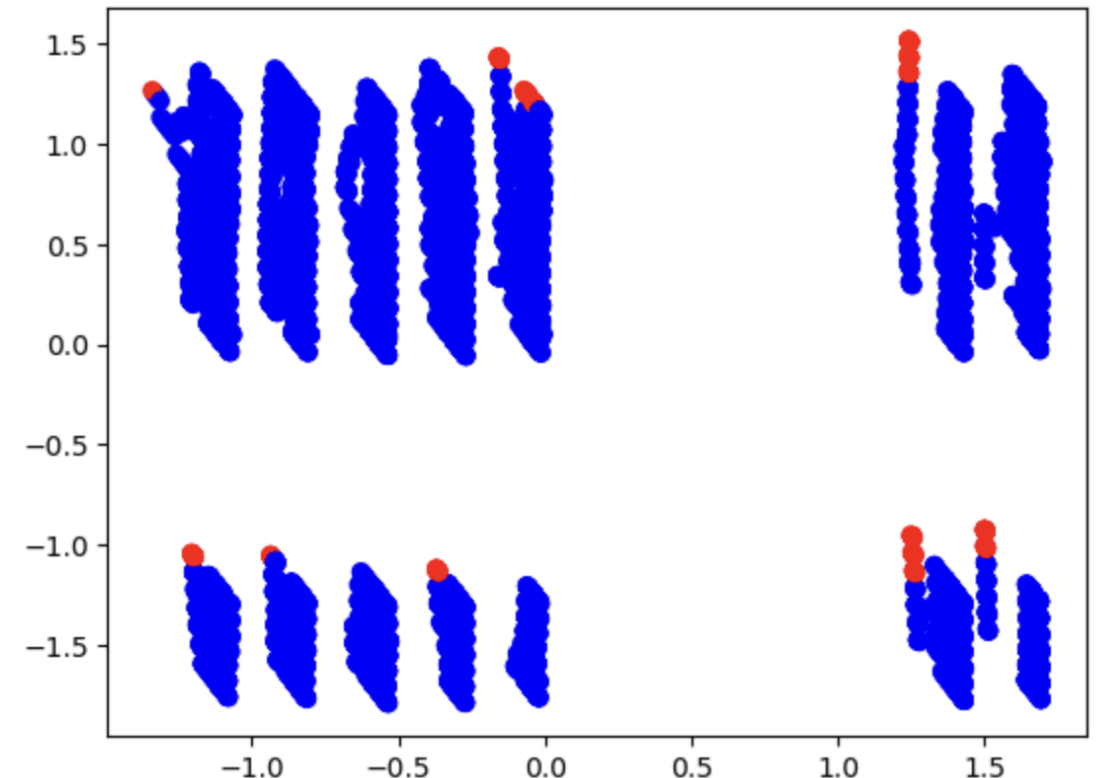


Fig.7 Anomaly points in clustering

Feature Engineering

Visualization of Anomaly Throughout Time

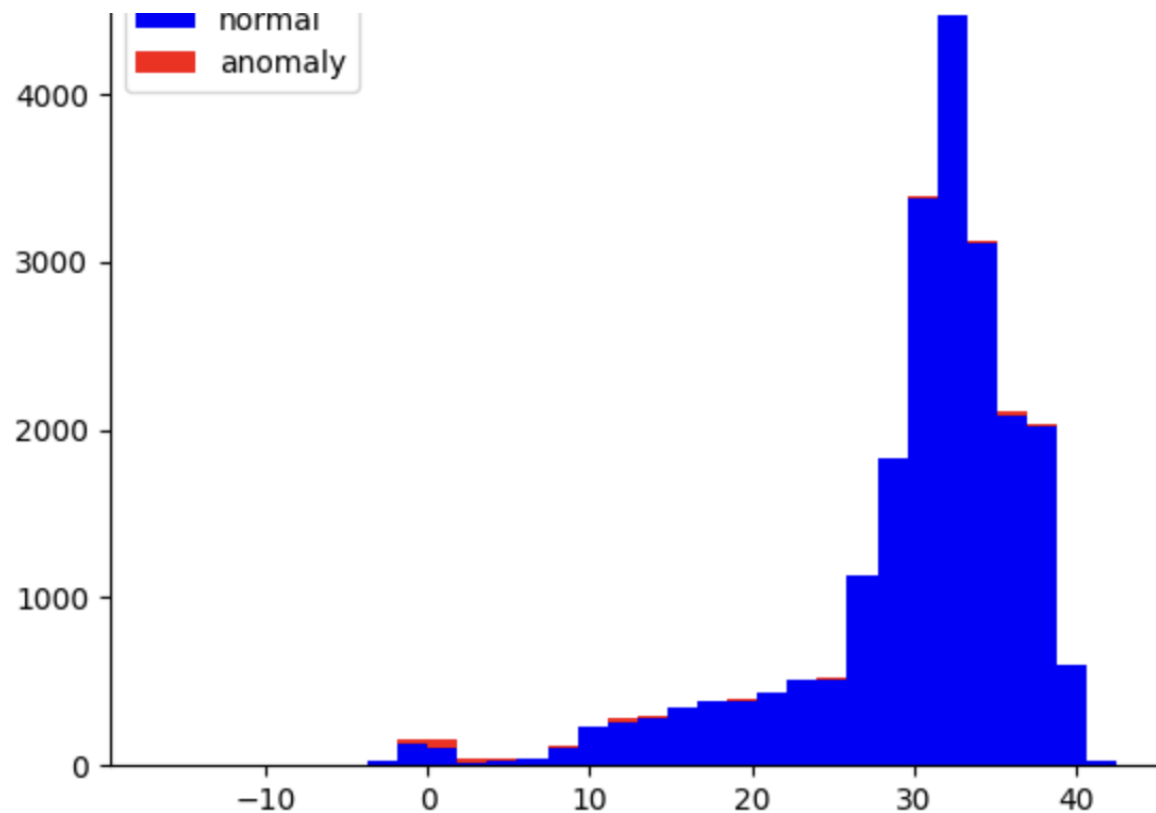


Fig.8 Anomaly with temperature repartition

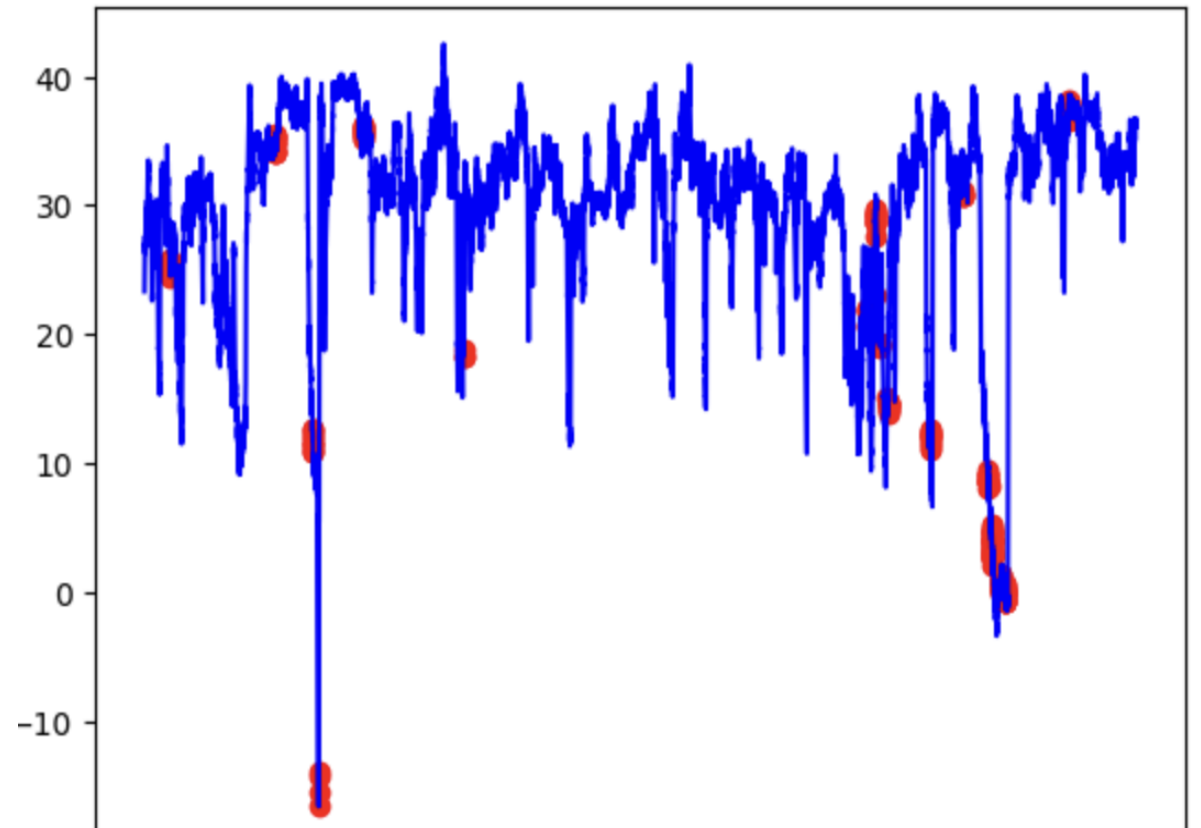


Fig.9 Anomaly throughout time

Models Used

Model	Purpose	Strength
Hotelling's T^2	Multivariate anomaly detection using statistical distance	Detects deviations from multivariate mean; effective for correlated features
One-Class SVM	Learns boundary around normal data to detect outliers	Works well with high-dimensional data; non-linear boundaries
Isolation Forest	Randomly partitions data to isolate anomalies	Fast, scalable; interprets anomalies as easier to isolate
Local Outlier Factor (LOF)	Detects local density deviations between neighbours	Captures context-based outliers; robust to local variations
ChangeFinder	Detects change points in time series using double AR modeling	Effective for sequential change detection in time series
Variance-Based Method	Flags anomalies based on sudden changes in rolling variance	Simple, interpretable; highlights volatility anomalies

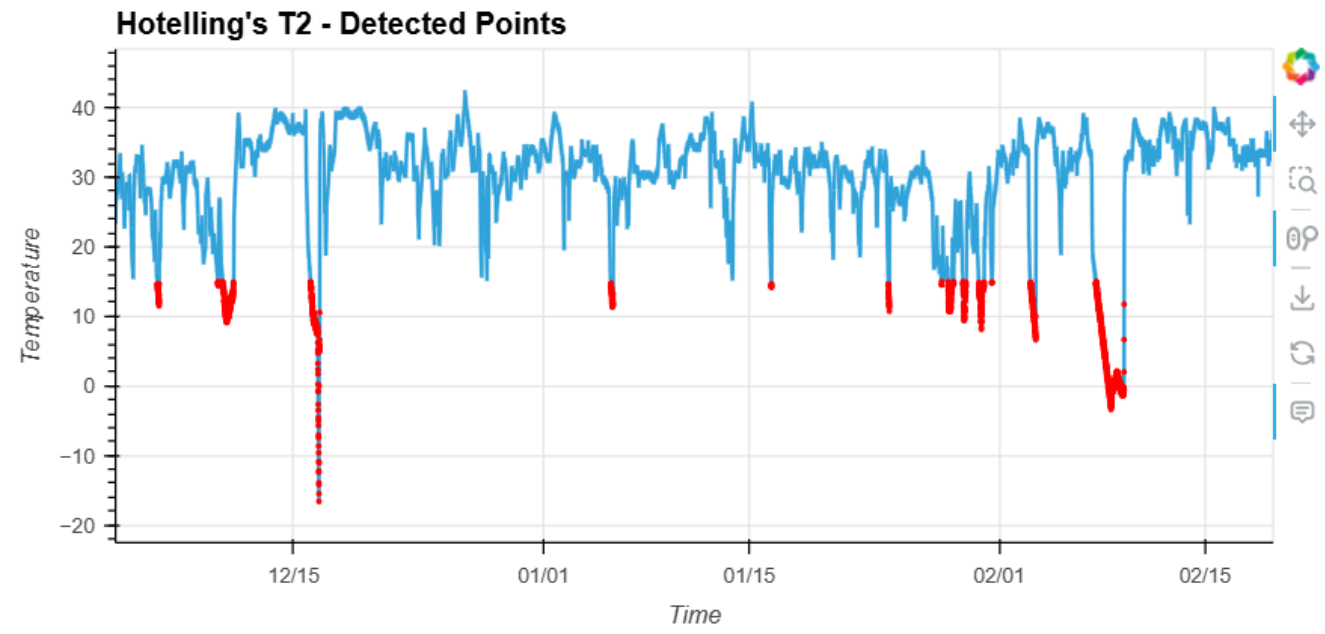
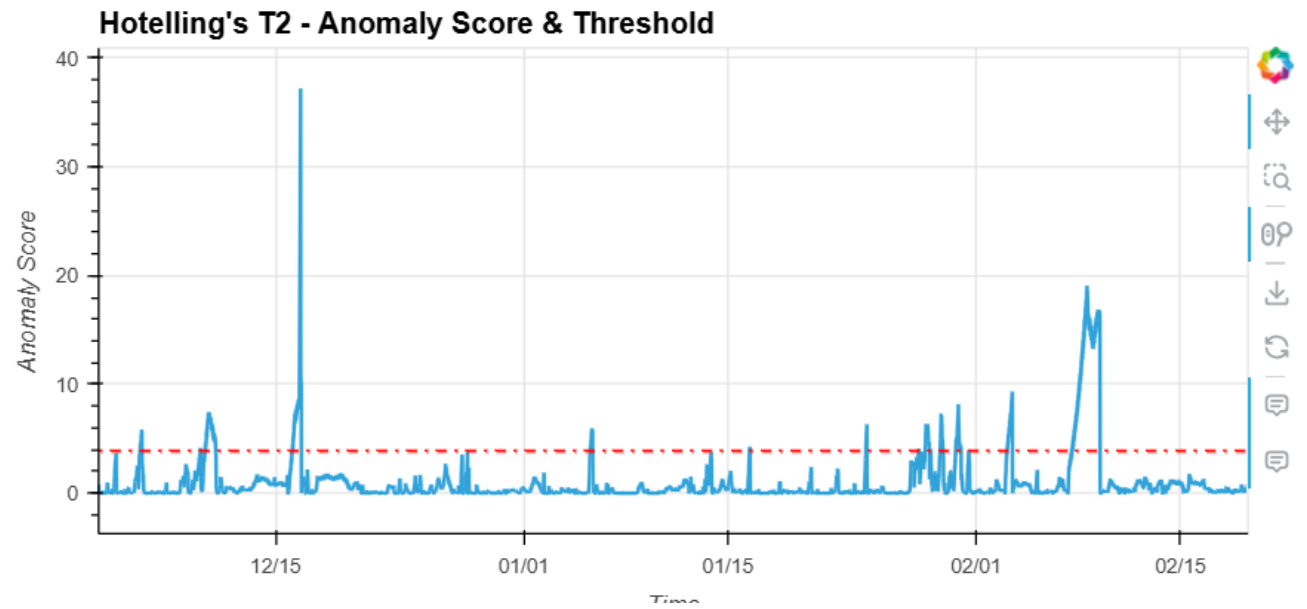
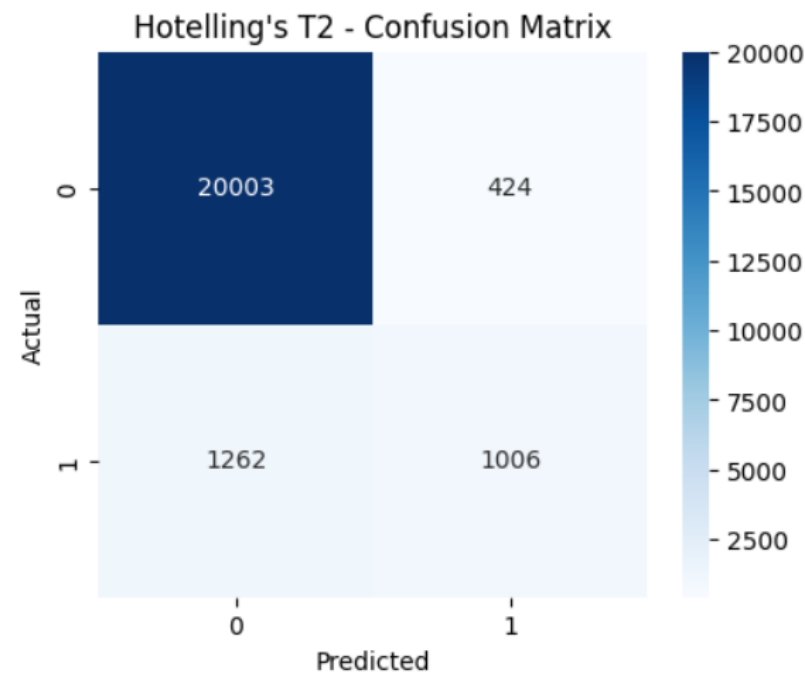
Table-4 Models used for Industrial Machine Anomaly Detection

Models' Results

Model-1: Hotellings's T2

Hotelling's T2 - Evaluation Metrics

Accuracy	: 0.9257
Precision	: 0.7035
Recall	: 0.4436
F1 Score	: 0.5441



Models' Results

Model-2: One-Class SVM

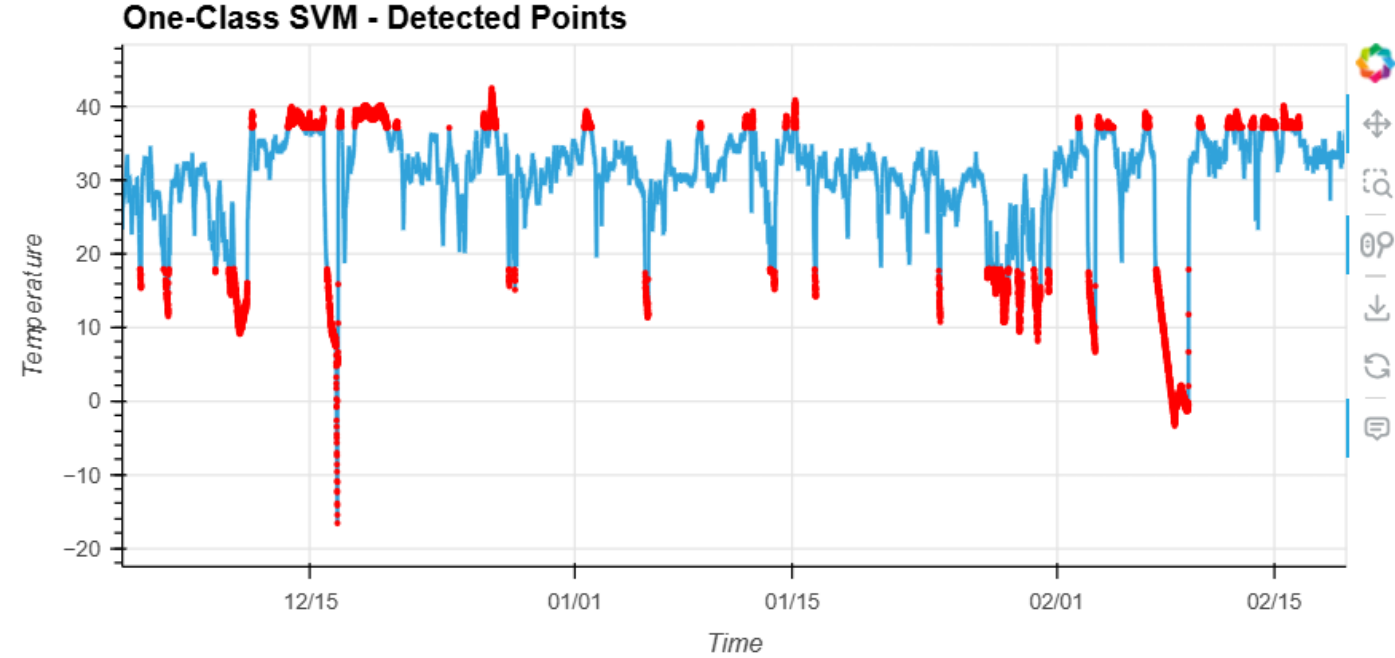
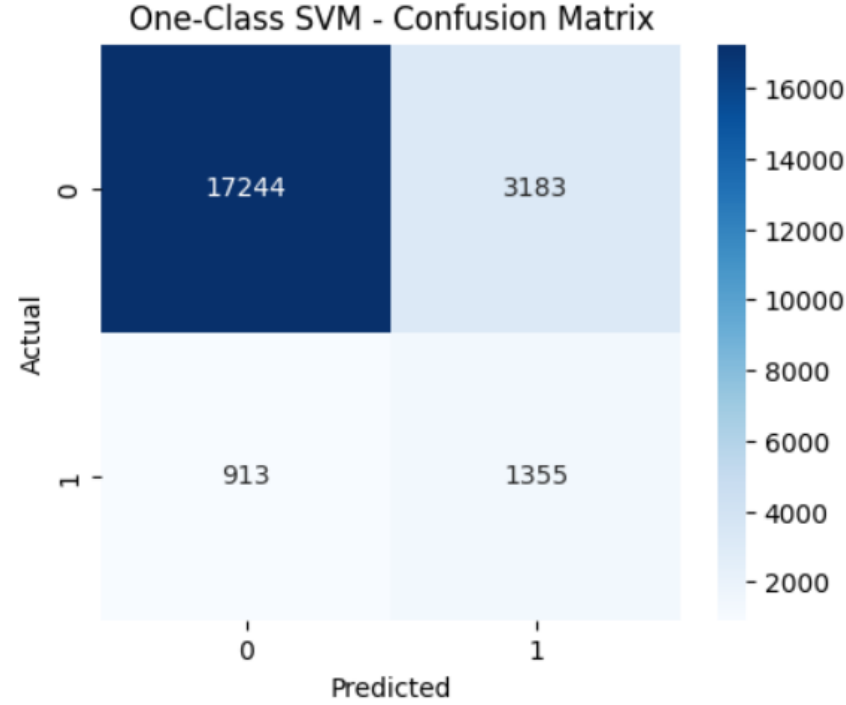
One-Class SVM - Evaluation Metrics

Accuracy : 0.8195

Precision : 0.2986

Recall : 0.5974

F1 Score : 0.3982



Models' Results

Model-3: Isolation Forest

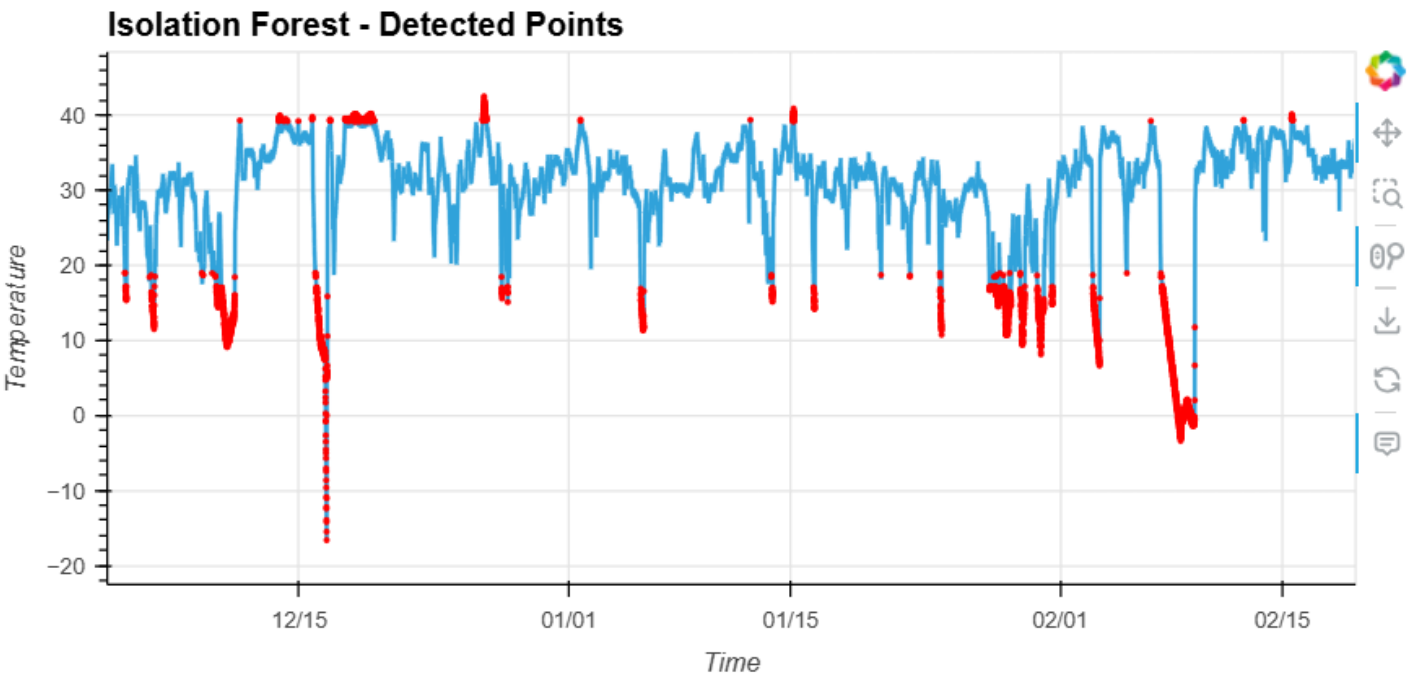
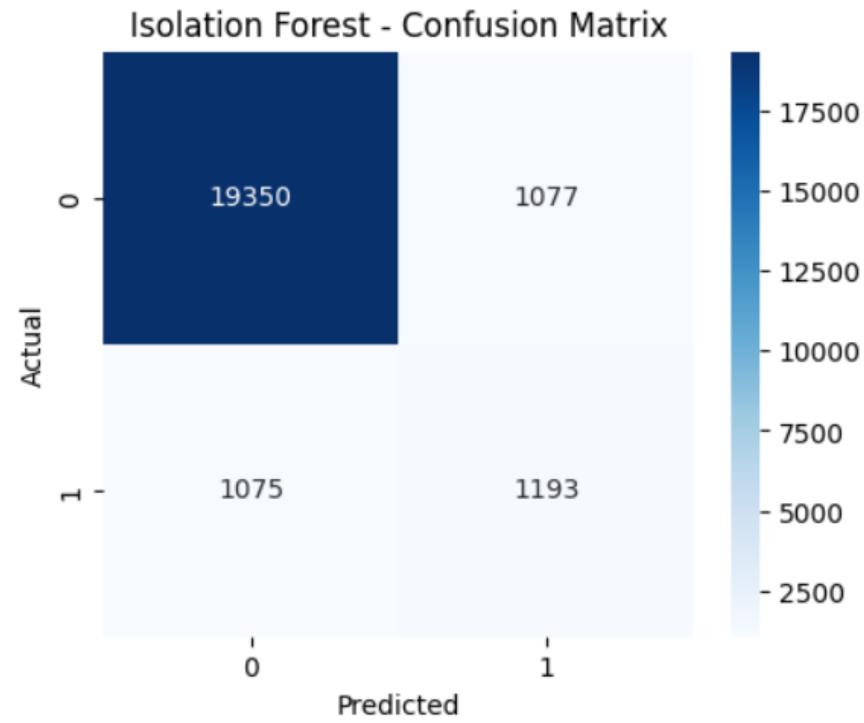
Isolation Forest - Evaluation Metrics

Accuracy : 0.9052

Precision : 0.5256

Recall : 0.5260

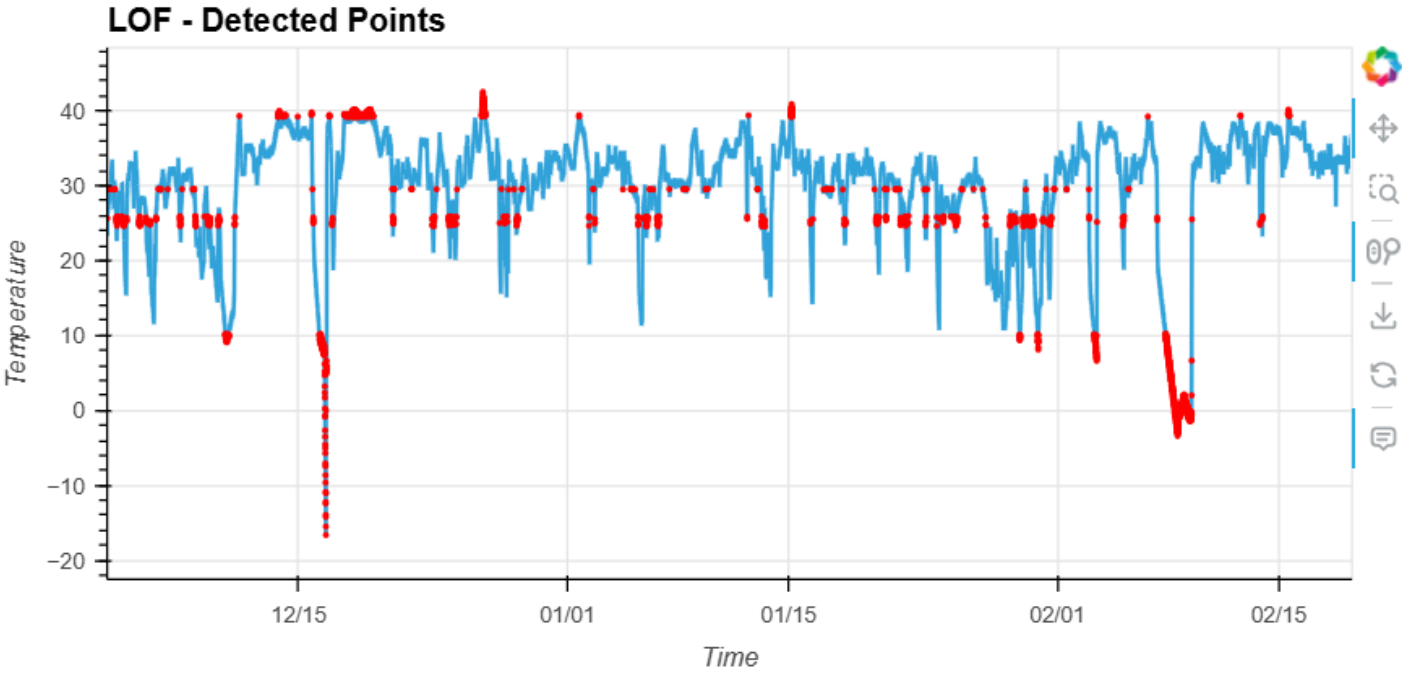
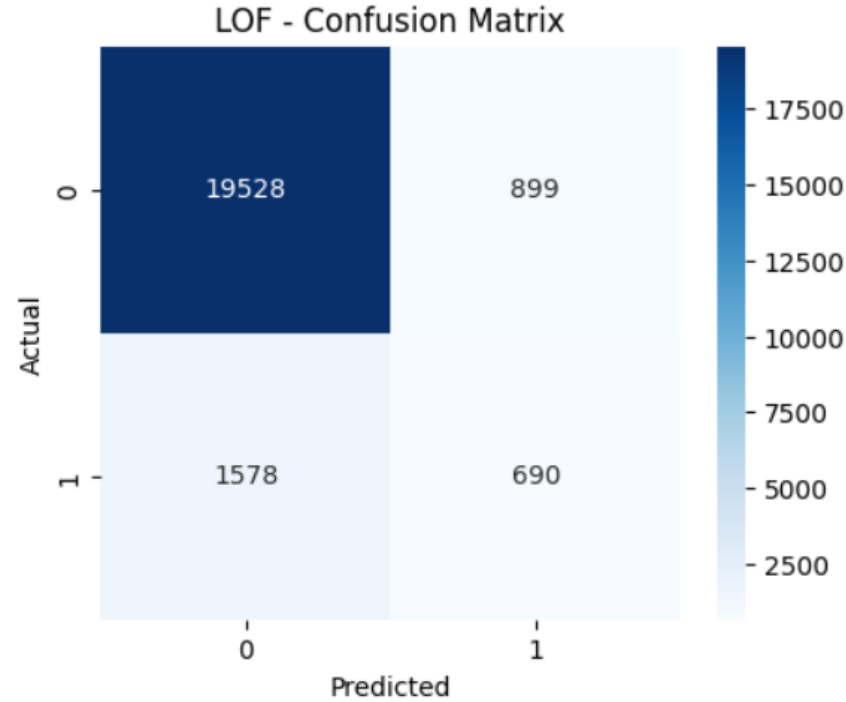
F1 Score : 0.5258



Models' Results

Model-4: LOF

LOF - Evaluation Metrics
Accuracy : 0.8909
Precision : 0.4342
Recall : 0.3042
F1 Score : 0.3578



Models' Results

Model-5: ChangeFinder

ChangeFinder - Evaluation Metrics

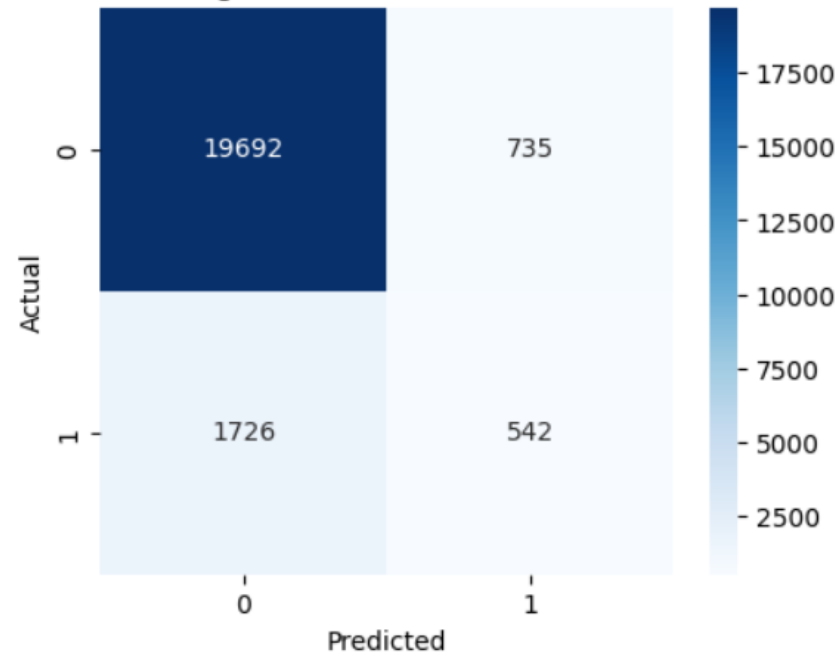
Accuracy : 0.8916

Precision : 0.4244

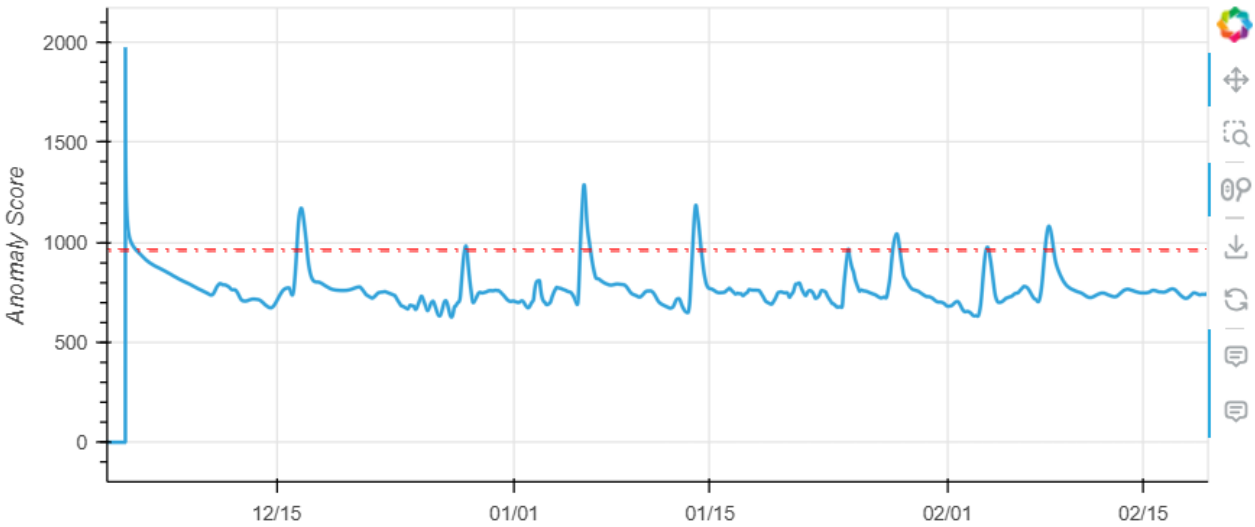
Recall : 0.2390

F1 Score : 0.3058

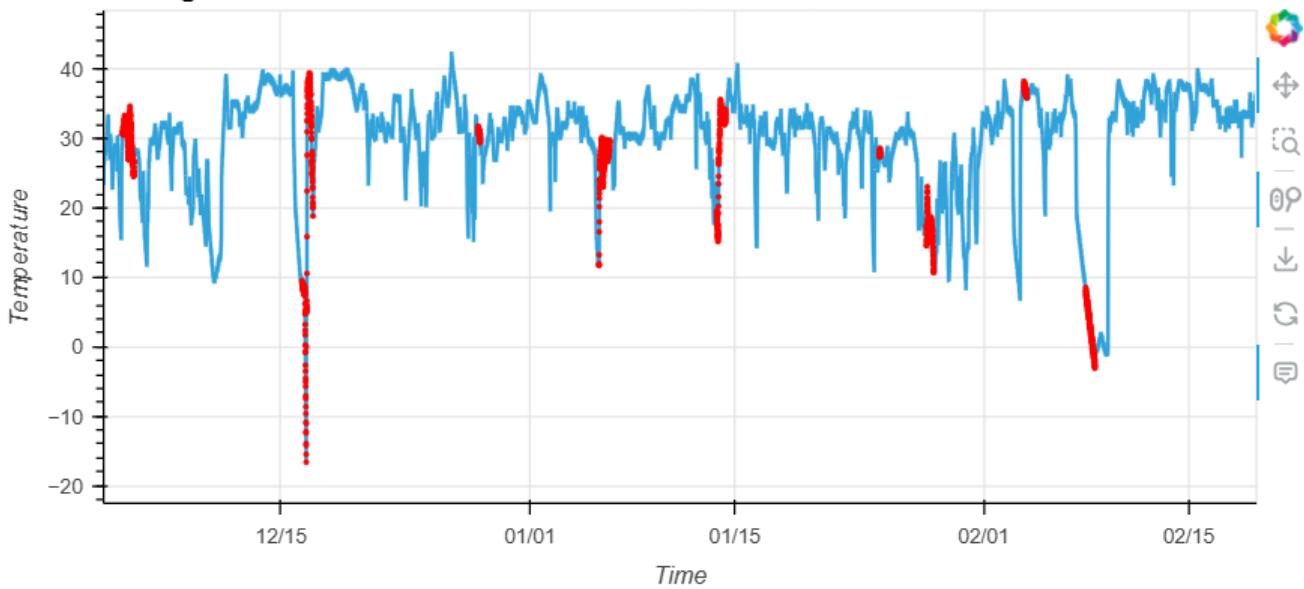
ChangeFinder - Confusion Matrix



ChangeFinder - Anomaly Score & Threshold



ChangeFinder - Detected Points

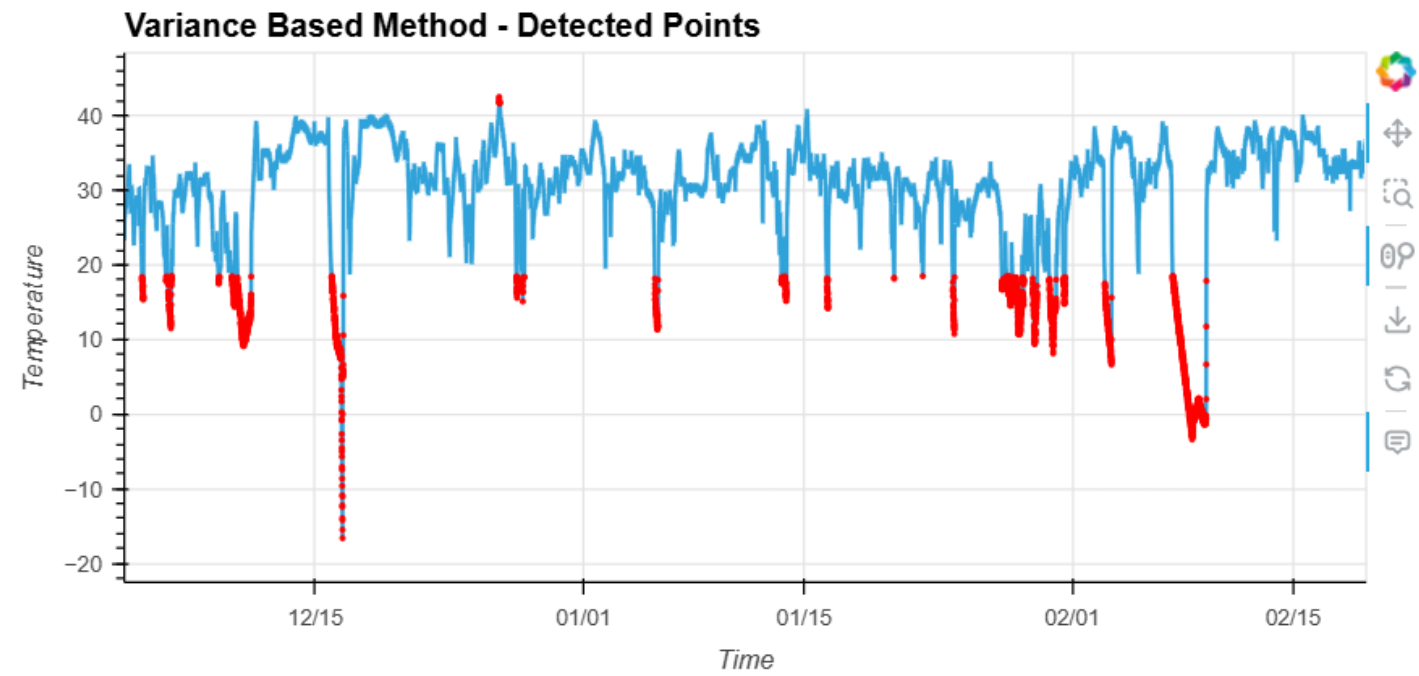
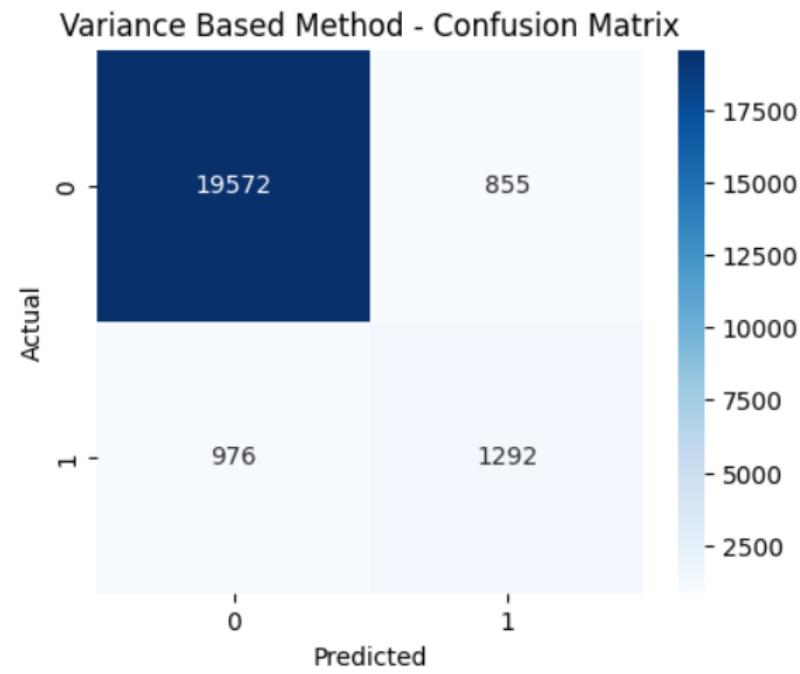


Models' Results

Model-6: Variance Based Method

Variance Based Method - Evaluation Metrics

Accuracy	: 0.9193
Precision	: 0.6018
Recall	: 0.5697
F1 Score	: 0.5853



Model Performance Comparison

	Accuracy	Precision	Recall	F1 Score
Model				
Hotelling's T2	0.925711	0.703497	0.443563	0.544078
One-Class SVM	0.819520	0.298590	0.597443	0.398178
Isolation Forest	0.905177	0.525551	0.526014	0.525782
LOF	0.890857	0.434235	0.304233	0.357791
ChangeFinder	0.891562	0.424432	0.238977	0.305783
Variance Based Method	0.919321	0.601770	0.569665	0.585277

Table-5 Performance Comparison of Models used for Anomaly Detection

Model Performance Comparison

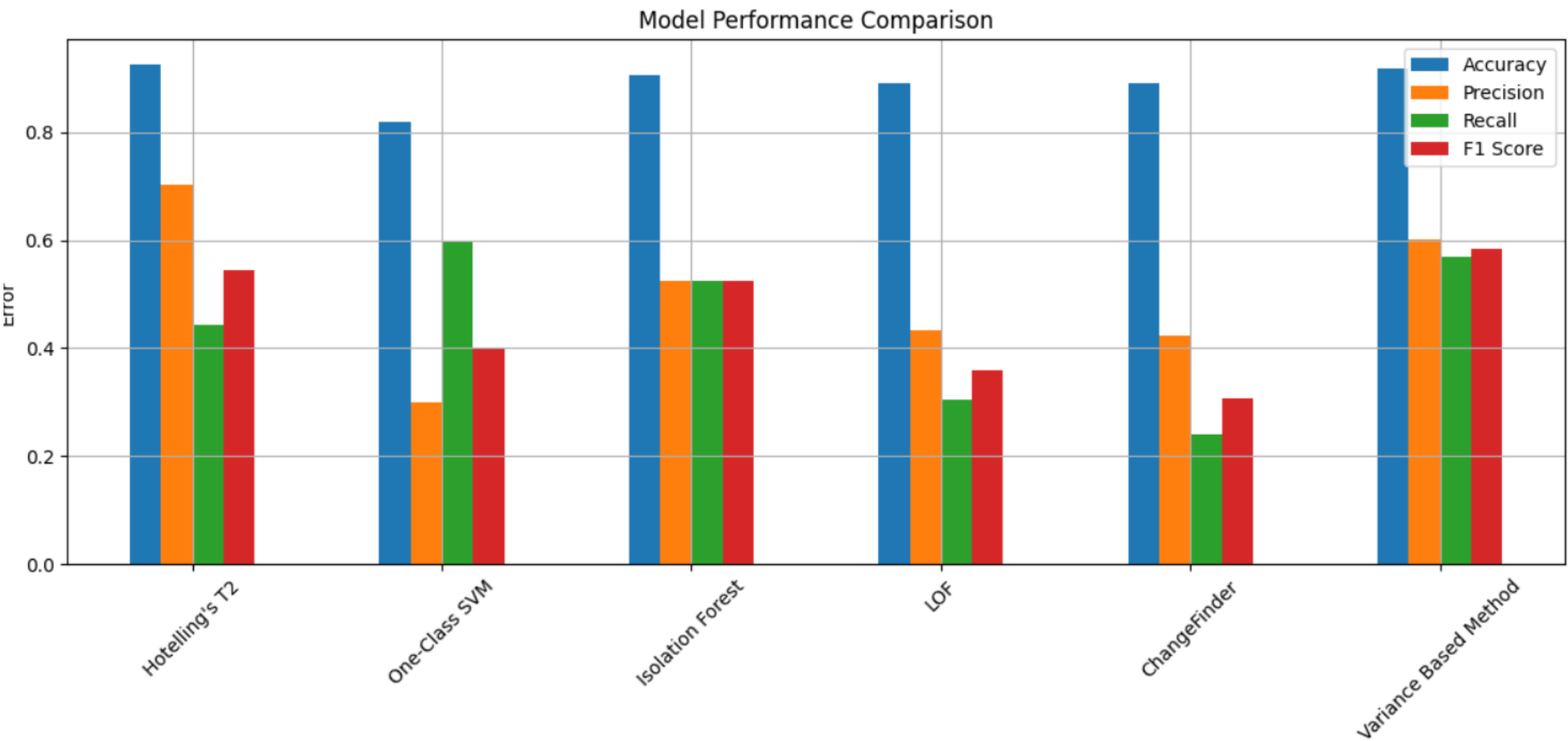


Fig.10 Model Comparison based on Evaluation Metrics

Model Performance Comparison

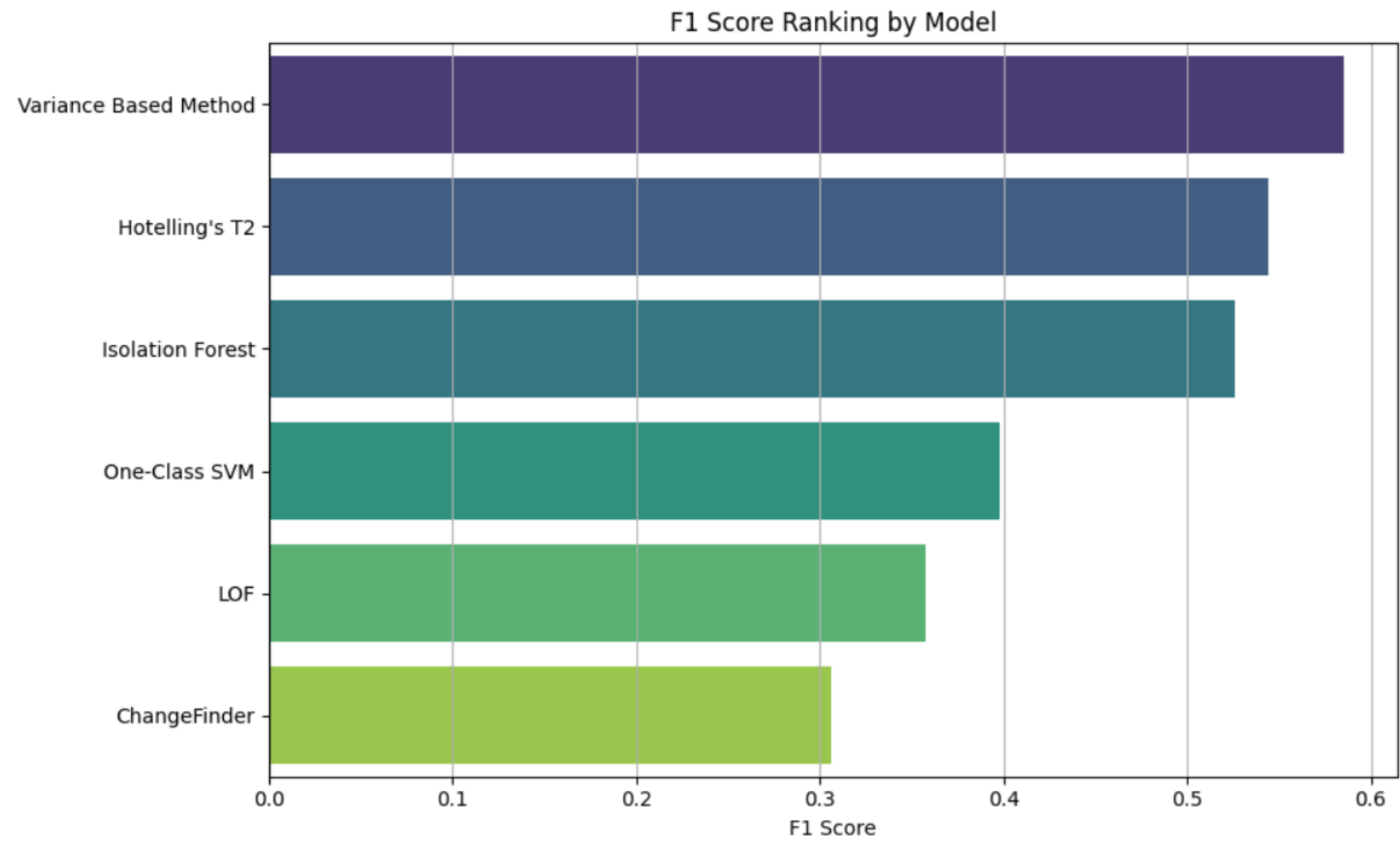


Fig.11 Model Comparison based on F1 Score

Model Performance Comparison

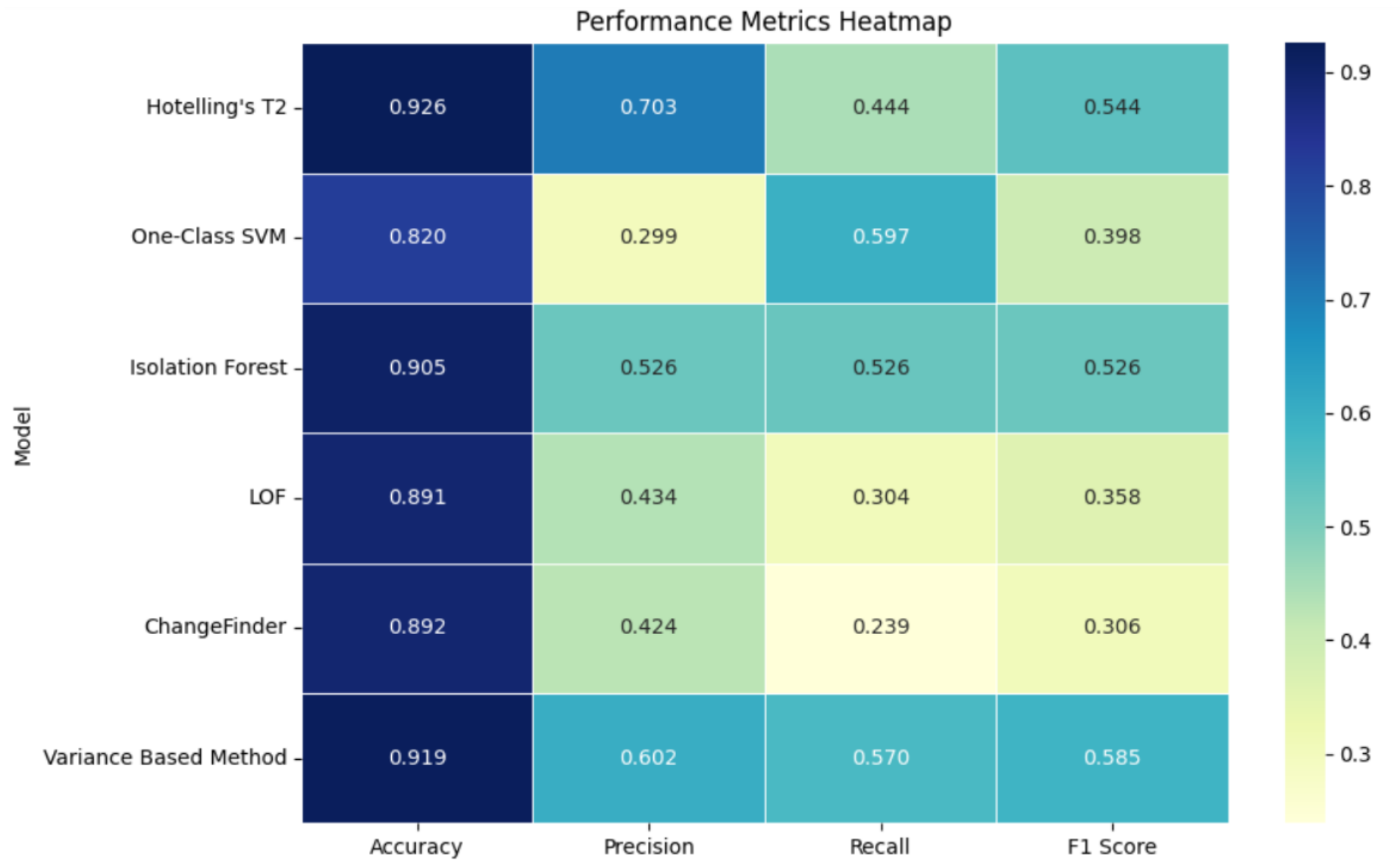


Fig.12 Model Comparison Heatmap

Key Findings

- **Variance-Based Method** had the highest F1 Score, indicating a strong balance between precision and recall. This suggests that the model was most effective at correctly identifying both actual anomalies and normal instances, minimizing false positives and false negatives.
- **Hotelling's T^2** had the highest accuracy and precision, making it great for minimizing false positives.
- **ChangeFinder** and **LOF** showed relatively low recall and F1 scores, suggesting they might be missing several anomalies.

Conclusion

- Simple algorithms (like Variance-Based Method or Hotelling's T^2) performed remarkably well and resulted in high accuracy.
- For time-series with a simple structure—like having a relatively constant mean and variance—simple methods are often sufficient and effective.
- On the other hand, when time-series become complex, with varying trends, seasonality, or multiple types of anomalies, then more sophisticated models (like Isolation Forest, ChangeFinder, One-Class SVM) are better suited.

References

- [1] Chevtchenko, S. F., Antunes, R. D. B., Schmitz, E. A., & Pereira, C. E. (2023). Anomaly Detection in Industrial Machinery using IoT Devices and Machine Learning: A Systematic Mapping. IEEE Access. <https://doi.org/10.1109/ACCESS.2023.32650188>
- [2] Umer, M., Imran, M., Habib, U., & Javaid, N. (2024). Machine Learning for Anomaly Detection in Industrial Environments. Computers in Industry. <https://doi.org/10.1016/j.compind.2024.1039411>
- [3] MathWorks. (2023). Anomaly Detection in Industrial Machinery Using Three-Axis Vibration Data. Retrieved from <https://www.mathworks.com/help/predmaint/ug/anomaly-detection-with-vibration-data.html>
- [4] Alam, M., Khan, R., & Khatri, A. (2024). Anomaly Detection in Industrial Sensors: A Machine Learning-Driven Approach to Predictive Maintenance. ResearchGate. <https://www.researchgate.net/publication/3739567544>
- [5] Nguyen, T., Dinh, D., & Phan, N. (2024). Explainable Correlation-Based Anomaly Detection for Industrial Control Systems. Frontiers in Artificial Intelligence, 7, 1225. <https://doi.org/10.3389/frai.2024.01225>
- [6] Numenta Anomaly Benchmark (NAB). (2016, August 19). <https://www.kaggle.com/datasets/boltzmannbrain/nab?resource=download>



Thank You